

Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-8574
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	React.js HTML5, CSS3, JavaScript	Responsive and interactive user interface for entering legal details, selecting document types, and previewing generated documents.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Node.js + Express.js	Handles API requests, user workflows, document generation logic, authentication, and communication with the AI service.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	MongoDB	Stores user profiles, document templates, generated-document metadata, and other flexible application data.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Render Docker	Provides simple and cost-effective deployment for the frontend/backend, with Docker supporting consistent application environments.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub GitHub Actions	Enables team collaboration, source-code versioning, and automated build/test/deployment workflows.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	OpenAI API PDF generation library	Uses AI to draft and refine legal-document content from user inputs and generates downloadable PDF documents.